

# Ada — Reference

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An **Ada** subset compiled with `lex` + `yacc` + `cc` (`ada.l` / `ada.y` lower Ada to C; `cc` lowers the C to .NET IL). The main procedure becomes `main`; every other subprogram (including ones nested in the main's declarative part) is hoisted to a `public static` method on `CProgram`, so compiled Ada is callable from C#/VB.NET.

Ada is **strongly, statically typed** and case-insensitive. It was deliberately designed to be LALR(1)-parseable, so it maps cleanly onto our toolchain.

```
ada prog.adb          # compile to prog.exe (native .NET executable)
ada prog.adb -o app.exe # choose the output name
ada lib.adb --dll      # compile to a library for C#/VB.NET to reference
```

## Quick Reference

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```
with Ada.Text_IO; use Ada.Text_IO;
procedure Main is
  X : Integer := 5;           -- variable with initializer
  Pi : constant Float := 3.14159; -- constant
  type Color is (Red, Green, Blue); -- enumeration
  A : array (1 .. 10) of Integer; -- array, indexed A(i), 1-based
  function Sq (N : Integer) return Integer is -- function (by value)
  begin return N * N; end Sq;
  procedure Swap (P : in out Integer; Q : in out Integer) is -- in out = by reference
  begin ... end Swap;
begin
  Put_Line ("text" & Integer'Image (X)); -- & concatenates; 'Image → string
  Put (Integer'Image (X)); New_Line;
  X := X + 1;          A (3) := 9;
  if C then ... elsif D then ... else ... end if;
  case N is when 1 ⇒ ... ; when 2 | 3 ⇒ ... ; when others ⇒ ... ; end case;
  for I in 1 .. 10 loop ... end loop;          for I in reverse 1 .. 10 loop ...
  while Cond loop ... end loop;                loop ... exit when Done; end loop;
end Main;
```

## Data types

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`Integer` (32-bit), `Float` (double), `Boolean` (`True` / `False`), `Character`, and `String`. **Enumerations** (`type Color is (Red, Green, Blue)`) are first-class — the literals are ordered constants. **Arrays** (`array (Lo .. Hi) of T`, named or anonymous) are indexed with parentheses and are 1-based by their lower bound. Numeric literals, `'c'` character literals, `" ... "` strings.

## Statements / Commands

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Assignment ( `:=` ), `if/elsif/else/end if`, `case ... when ... when others ... end case`, the loop forms ( `loop`, `while ... loop`, `for I in [reverse] L .. H loop` ) with `exit` / `exit when`, procedure calls, `return`, `null;`, and `Ada.Text_IO` output.

## Functions

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Two kinds of subprogram. A **function** returns a value ( `return expr;` ) and a **procedure** returns through its parameters or not at all. Parameter modes: `in` (the default) passes **by value**; `out` and `in out` pass **by reference**, so `Swap (X, Y)` exchanges the caller's variables. Each subprogram compiles to a `public static` method `ada_<name>` on `CProgram` — the basis for C#/VB.NET interop (Activity 9).

## Input / Output

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Via `Ada.Text_IO`: `Put_Line (S)` writes a string and a newline, `Put (S)` without the newline, `New_Line` ends a line. Numbers are turned into strings with the `'Image` attribute ( `Integer'Image (X)`, `Float'Image (X)` ) — note Ada's `'Image` puts a leading space before a non-negative number, which this subset reproduces.

## Graphics

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None. Ada is general-purpose/systems-oriented; Activity 8 below builds a text histogram.

## Notes

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- **Case-insensitive:** `Put_Line`, `put_line`, and `PUT_LINE` are the same.
- `=` is equality, `≠` is inequality, `:=` is assignment, `&` concatenates strings.
- `and then` / `or else` are the short-circuit forms; `mod` / `rem` / `abs` / `**` are operators.
- The `'` tick is an attribute marker ( `X'Image` ) except where a character literal ( `'a'` ) is expected — disambiguated by context in the scanner.

## Subset boundaries

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A solid procedural core. Not included: packages with separate spec/body ( `package ... is` ), generics, tasking, exceptions, `record` types, access (pointer) types, `declare` blocks, 2-D arrays and array slices/attributes ( `'First` / `'Last` / `'Length` ), and the wider standard library. Nested subprograms are *hoisted* (they don't capture the enclosing procedure's locals). Numbers are 32-bit `Integer` / 64-bit `Float`.

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# Tutorial

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Every example was compiled and run with `ada`; the output shown is real.

## 1. Your first program

```
with Ada.Text_IO; use Ada.Text_IO;
procedure Main is
begin
    Put_Line ("Hello from Ada");
end Main;
```

```
Hello from Ada
```

`with / use` make `Ada.Text_IO` visible; `procedure Main` is the entry point and becomes `main`.

## 2. Variables and data types

```
X      : Integer := 5;
Name   : String  := "Ada";
Ok      : Boolean := True;
type Color is (Red, Green, Blue);
C       : Color  := Green;
```

Each object is declared with a type after `:`. Enumerations like `Color` give named, ordered values (`Green` is the constant 1).

## 3. Flow control

```
if N > 10 then
    Put_Line ("big");
elsif N > 0 then
    Put_Line ("small");
else
    Put_Line ("zero");
end if;

case C is
    when Red    => Put_Line ("red");
    when Green => Put_Line ("green");
    when Blue  => Put_Line ("blue");
end case;
```

With `N = 5` and `C = Green`:

```
small  
green
```

`case ... when` is the multi-way branch ( `when others =>` is the catch-all); `elsif` chains conditions.

## 4. Arrays

Arrays are 1-based and indexed with parentheses:

```
A : array (1 .. 5) of Integer;  
Sum : Integer := 0;  
...  
for I in 1 .. 5 loop  
    A (I) := I * I;  
end loop;  
for I in reverse 1 .. 5 loop  
    Sum := Sum + A (I);  
end loop;  
Put_Line ("Squares sum 1..5 =" & Integer'Image (Sum));
```

```
Squares sum 1..5 = 55
```

`for I in reverse 1 .. 5` counts down; `A (I)` indexes the array.

## 5. Subroutines and functions

A **function** returns a value; a **procedure** with `in out` parameters mutates the caller's variables:

```
function Square (N : Integer) return Integer is  
begin  
    return N * N;  
end Square;  
  
procedure Swap (A : in out Integer; B : in out Integer) is  
    T : Integer;  
begin  
    T := A; A := B; B := T;  
end Swap;
```

With `X = 5`: `Integer'Image (Square (X))` gives `25`, and after `Swap (X, Total)` the two variables are exchanged — exactly Ada's `in out` by-reference semantics.

## 6. Memory management

Storage is **static / stack** — there is nothing to allocate or free:

- Each variable is a fixed object for the lifetime of its subprogram (or the program, for the main's variables).
- `array (1 .. N) of T` reserves `N` elements; `constant` makes a read-only value.
- Subprogram parameters and locals live for the duration of the call.

Heap allocation (access types) is outside this subset.

## 7. Strings and enumerations

Strings are joined with `&`, and `'Image` turns any scalar into its textual form:

```
type Day is (Mon, Tue, Wed);
D : Day := Wed;
N : Integer := 42;
Put_Line ("Day" & Integer'Image (D) & ", N =" & Integer'Image (N));
```

```
Day 2, N = 42
```

`'Image` of an enumeration gives its position (`Wed` = 2); concatenation builds the line.

## 8. Drawing a picture — a text histogram

```
with Ada.Text_IO; use Ada.Text_IO;
procedure Main is
  Data : array (1 .. 4) of Integer;
  Bar : String := "";
begin
  Data (1) := 3; Data (2) := 6; Data (3) := 2; Data (4) := 5;
  for I in 1 .. 4 loop
    Bar := "";
    for J in 1 .. Data (I) loop
      Bar := Bar & "#";
    end loop;
    Put_Line (Bar);
  end loop;
end Main;
```

```
###
#####
##
#####
```

Each row concatenates `Data (I)` hashes — a histogram drawn with `&` and a nested loop.

## 9. Calling Ada from C# / VB.NET

Compile an Ada file of subprograms as a library; each becomes a `public static` method on `CProgram`:

```
function Add (A : Integer; B : Integer) return Integer is
begin
    return A + B;
end Add;

function Fib (N : Integer) return Integer is
    A : Integer := 0; B : Integer := 1; T : Integer;
begin
    for I in 1 .. N loop
        T := A + B; A := B; B := T;
    end loop;
    return A;
end Fib;
```

```
ada lib.adb --dll          # → adalib.dll
```

A C# program referencing `adalib.dll` (and `CRuntime.dll`) calls them directly:

```
Console.WriteLine($"add(20, 22) = {CProgram.ada_add(20, 22)}");
Console.WriteLine($"fib(15)      = {CProgram.ada_fib(15)}");
```

```
add(20, 22) = 42
fib(15)     = 610
```

Because parameters are by value, the signatures are ordinary .NET ones ( `int ada_add(int, int)` ) — full type checking on the C# side. From here the natural extensions are records, packages, and exceptions (*Subset boundaries*).